

Experiment - 1.1.1. Area of Circle

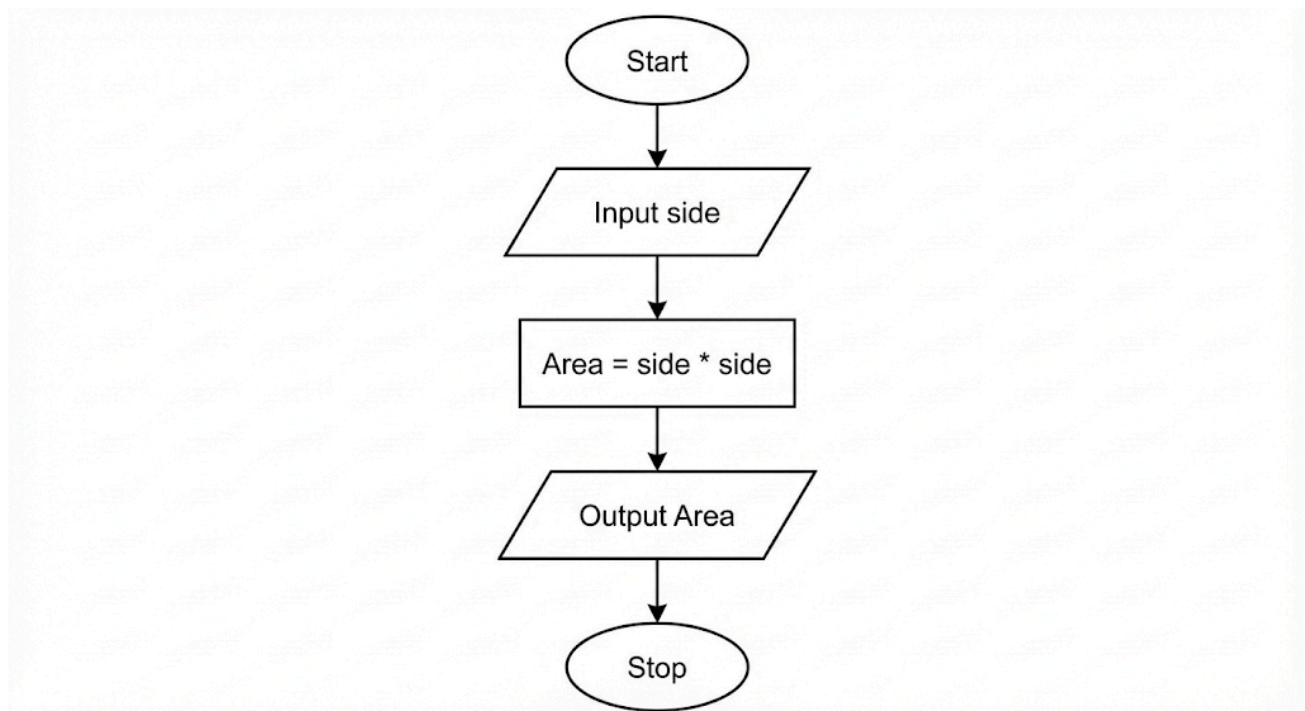
1. Aim

To write a Python program that calculates the area of a circle given its radius as input from the user, using the constant value of $\pi = 3.14$ and displaying the result formatted to four decimal places.

2. Pseudocode

1. **START**
2. **READ** the input value from the user and convert it to a floating-point number.
3. **STORE** the value in a variable named radius.
4. **DEFINE** a constant variable pi and assign it the value 3.14.
5. **CALCULATE** the area using the formula: $Area = \pi \times radius \times radius$.
6. **FORMAT** the resulting area to 4 decimal places.
7. **PRINT** the formatted area.
8. **END**

3. Flowchart



4. Python Program

```
# Program to calculate the area of a circle
# Input: Radius as a float
# Output: Area formatted to 4 decimal places
```

```
# Taking input from the user
radius = float(input())
```

```
# Constant value of Pi
pi = 3.14
```

```
# Calculating area
area = pi * radius * radius
```

```
# Displaying the result formatted to 4 decimal places
print(f"{area:.4f}")
```

5. Experiment Screenshot

The screenshot displays the CODETANTRA online IDE interface. On the left, the problem statement for "1.1.1. Area of Circle" is shown, along with input and output formats and sample test cases. The main editor shows the Python code for calculating the area of a circle. The code is as follows:

```
1 radius = float(input())
2
3 pi = 3.14
4 area = pi * radius * radius
5
6 print(f"{area:.4f}")
7
8
```

Below the code editor, the execution results are displayed, showing that 2 out of 2 shown test case(s) passed and 2 out of 2 hidden test case(s) passed. The average time is 0.006 s and the maximum time is 0.010 s. The test case details for Test case 1 are shown, with an expected output of 35.4493 and an actual output of 35.4493.

At the bottom of the interface, there are buttons for "Terminal", "Test cases", "Prev", "Reset", "Submit", and "Next".

Experiment - 1.1.2. Area of Rectangle

1. Aim

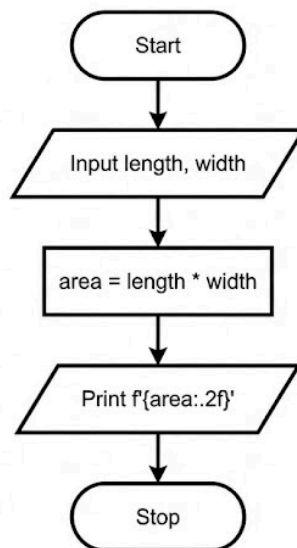
To design and implement a Python program that calculates the area of a rectangle. The program accepts the length and width as floating-point inputs from the user and computes

the area using the formula $Area = Length \times Width$, displaying the result formatted to two decimal places.

2. Pseudocode

1. **START**
2. **READ** the first input value and store it as a float in the variable length.
3. **READ** the second input value and store it as a float in the variable width.
4. **CALCULATE** the area by multiplying length and width.
5. **STORE** the result in the variable area.
6. **FORMAT** the value of area to 2 decimal places.
7. **PRINT** the formatted result.
8. **END**

3. Flowchart



4. Python Program

Program to calculate the area of a rectangle
Input: Length and Width as floats
Output: Area formatted to 2 decimal places

Taking inputs from the user
length = float(input())
width = float(input())

Calculating area
area = length * width

Displaying the result formatted to 2 decimal places
print(f"area:.2f")

5. Experiment Screenshot

The screenshot displays the CODETANTRA IDE interface. On the left, a sidebar shows the file explorer with a file named 'areaOfRe...'. The main editor area contains the following Python code:

```
1 length = float(input())
2 width = float(input())
3
4 area = length * width
5
6 print(f"area:.2f")
7
8
```

Below the code editor, the test results are shown. The average time is 0.004 s (4.30 ms) and the maximum time is 0.007 s (7.00 ms). The test results indicate that 5 out of 5 shown test case(s) passed and 5 out of 5 hidden test case(s) passed. A detailed view of Test case 1 shows the expected output as 10.5 and 5.2, and the actual output as 10.5 and 5.2.

The bottom of the interface features a terminal and a 'Test cases' button. The footer includes navigation buttons: < Prev, Reset, Submit, and Next >.

Experiment - 1.1.3. Calculate Area of the Square

1. Aim

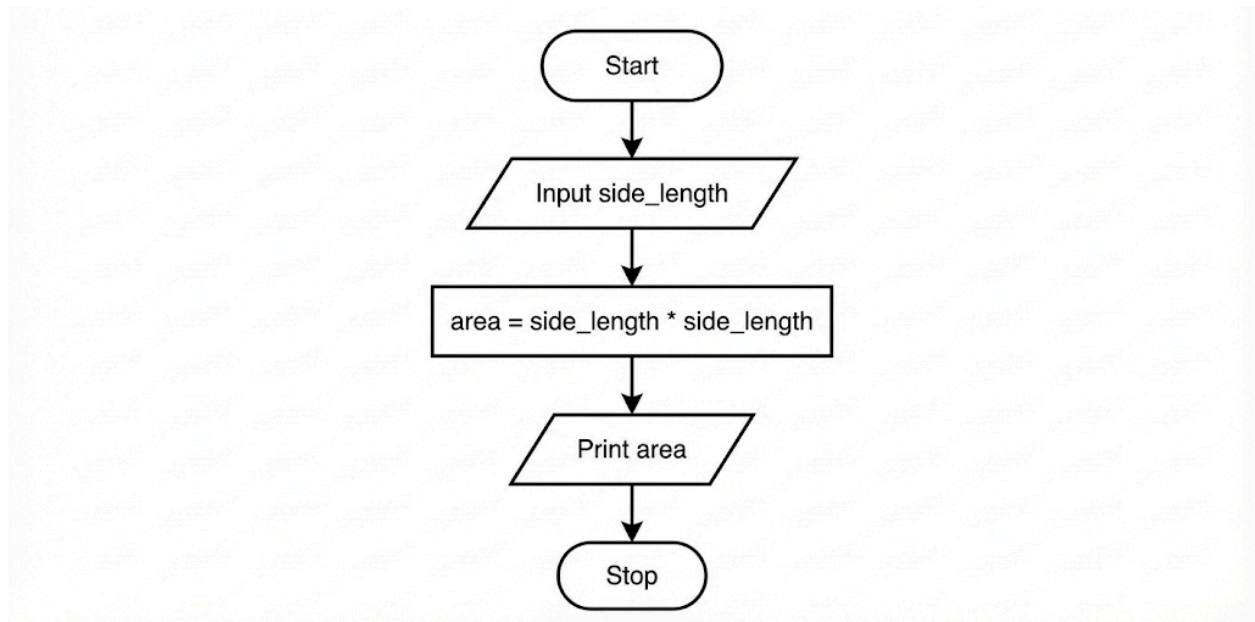
To design and implement a Python program that calculates the area of a square. The program accepts the `side_length` as an integer input and computes the area using the formula

$$\text{Area} = \text{side_length}^2$$

2. Pseudocode

1. **START**
2. **READ** the input value from the user and convert it to an integer.
3. **STORE** this value in the variable `side_length`.
4. **CALCULATE** the area by multiplying `side_length` by itself (`side_length * side_length`).
5. **STORE** the result in the variable `area`.
6. **PRINT** the value of `area`.
7. **END**

3. Flowchart



4. Python Program

```
# Program to calculate the area of a square
# Input: side_length as an integer
```

```
# Output: area as an integer
```

```
# Taking input from the user  
side_length = int(input())
```

```
# Calculating area (side * side)  
area = side_length * side_length
```

```
# Displaying the result  
print(area)
```

5. Experiment Screenshot

The screenshot displays the CODETANTRA online IDE interface. On the left, the problem description for "1.1.3. Calculate Area of the Square" is shown, including the formula $\text{Area} = \text{side_length}^2$, input/output formats, and sample test cases. The main editor shows a Python program that takes an input for `side_length` and prints the calculated area. The test results panel on the right indicates that 2 out of 2 shown test cases passed, with a detailed view of Test case 1 showing an expected output of 25 and an actual output of 25.

CODETANTRA Home saksham.wadhankar.batch2025@sitnagpur.siu.edu.in Support Logout

1.1.3. Calculate Area of the Square 0025

Write a Python program that prompts the user to enter the *side_length* of a square and computes the area of the square.

Formula:

- Area = side_length^2

Input Format:

- The input is a positive integer value that represents the *side_length* of the square.

Output Format:

- The output is a positive integer value that represents the area of the square.

Sample Test Cases +

AreaSqua... Submit

```
1 side_length = int(input())  
2  
3 area = side_length * side_length  
4  
5 print(area)  
6  
7
```

Average time: 0.003 s (3.00 ms) Maximum time: 0.004 s (4.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output: 25
Actual output: 25

Terminal Test cases

< Prev Reset Submit Next >

Experiment - 1.1.4. Area of Triangle

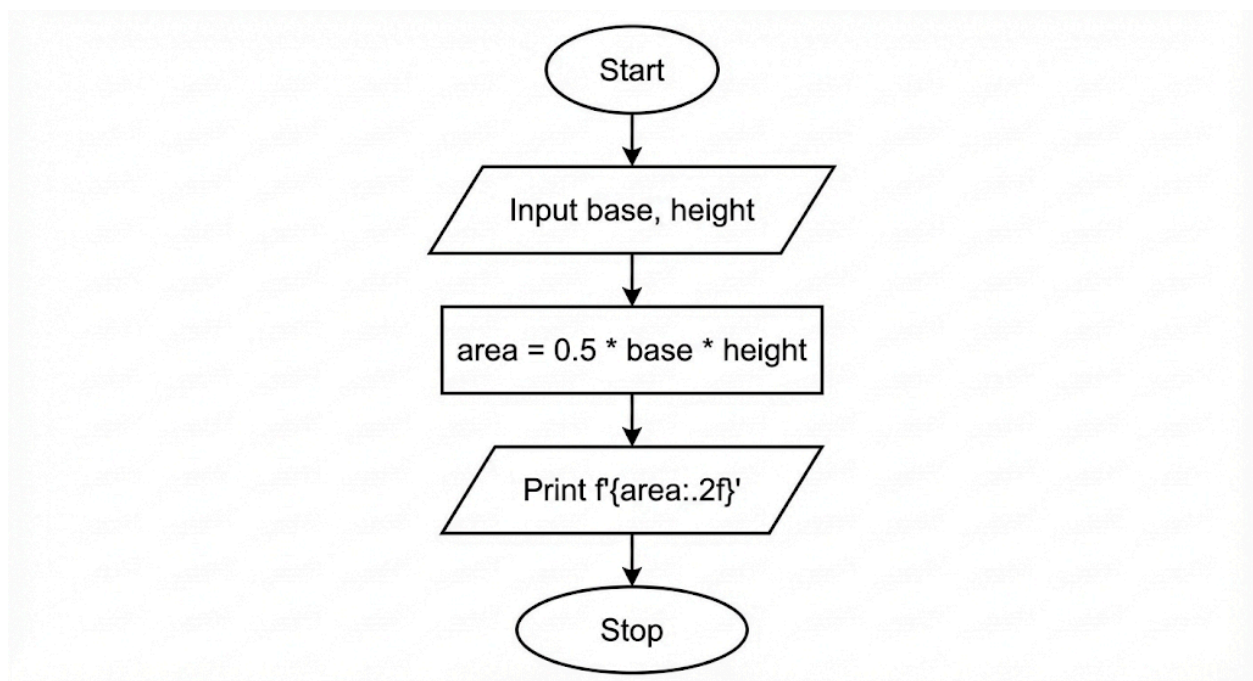
1. Aim

To design and implement a Python program that calculates the area of a triangle. The program prompts the user to enter the base and height as floating-point values and computes the area using the formula $Area = 0.5 \times base \times height$, with the result formatted to two decimal places.

2. Pseudocode

1. **START**
2. **READ** the first input value and store it as a float in the variable base.
3. **READ** the second input value and store it as a float in the variable height.
4. **CALCULATE** the area using the formula: $0.5 * base * height$.
5. **STORE** the result in the variable area.
6. **FORMAT** the result to show 2 decimal places.
7. **PRINT** the formatted area.
8. **END**

3. Flowchart



4. Python Program

Program to calculate the area of a triangle

Input: Base and Height as float values
Output: Area formatted to two decimal places

Taking inputs from user
base = float(input())
height = float(input())

Calculating the area
area = 0.5 * base * height

Displaying the output formatted to 2 decimal places
print(f"{area:.2f}")

5. Experiment Screenshot

The screenshot displays the CODETANTRA IDE interface. On the left, the problem description for "1.1.4. Area of Triangle" is shown, including the formula $\text{Area of Triangle} = 0.5 \times \text{base} \times \text{height}$ and input/output specifications. The main editor shows the following Python code:

```
1 base = float(input())
2 height = float(input())
3
4 area = 0.5 * base * height
5
6 print(f"{area:.2f}")
7
8
```

Below the code editor, the test runner shows the following results:

- Average time: 0.006 s (5.75 ms)
- Maximum time: 0.009 s (9.00 ms)
- 2 out of 2 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

Test case 1 details:

Expected output	Actual output
6.54	6.54
1.23	1.23

The interface also includes a "Sample Test Cases" input field and navigation buttons at the bottom: Prev, Reset, Submit, and Next.

Experiment - 1.1.5. Student Pass or Fail Status

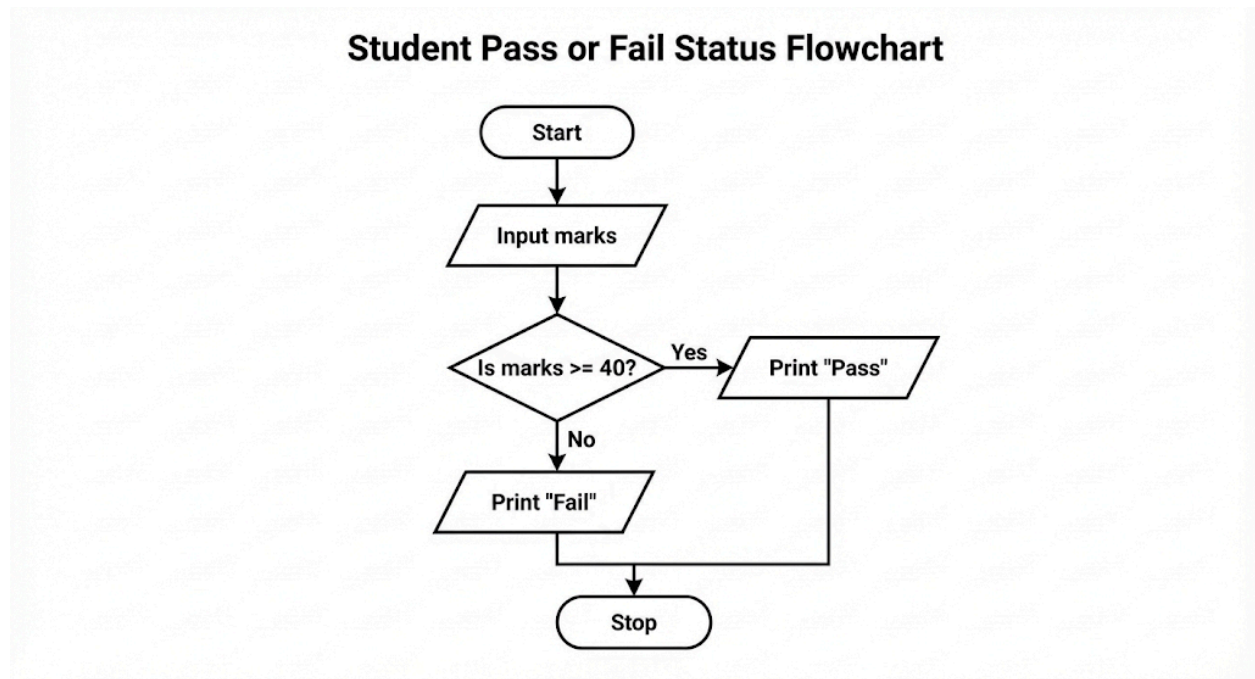
1. Aim

To design and implement a Python program that determines a student's examination status based on their marks. The program evaluates whether the student has passed or failed using a threshold of 40 marks (Pass: $marks \geq 40$, Fail: $marks < 40$).

2. Pseudocode

1. **START**
2. **READ** the input value from the user and convert it to an integer.
3. **STORE** the value in the variable marks.
4. **IF** marks is greater than or equal to 40:
 - **PRINT** "Pass"
5. **ELSE:**
 - **PRINT** "Fail"
6. **END**

3. Flowchart



4. Python Program

```
# Program to determine student pass/fail status
# Input: marks as an integer
```

```
# Output: "Pass" or "Fail" based on criteria
```

```
# Taking marks as input
```

```
marks = int(input())
```

```
# Conditional check for passing criteria
```

```
if marks >= 40:
```

```
    print("Pass")
```

```
else:
```

```
    print("Fail")
```

5. Experiment Screenshot

The screenshot displays the CodeTANTRA IDE interface. On the left, a sidebar shows the problem statement: "1.1.5. Student Pass or Fail Status". The problem asks to write a Python program to determine if a student passed or failed based on their marks. It specifies the pass/fail criteria: a student passes if marks are ≥ 40 and fails if marks are < 40 . It also defines the input and output formats. The main area shows the Python code for this logic. The code is as follows:

```
1 marks = int(input())
2
3 if marks >= 40:
4     print("Pass")
5 else:
6     print("Fail")
7
8
```

Below the code editor, the test results are shown. The average time is 0.003 s and the maximum time is 0.004 s. The results indicate that 3 out of 3 shown test cases passed and 4 out of 4 hidden test cases passed. A detailed view of Test case 1 shows the expected output "Pass" and the actual output "Pass", both for an input of 45.